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Clankers in the Cultural Imagination: Online Robophobia and Its Implications for Human-Robot Interaction

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Abstract

Robophobia is a recent and growing trend on social media, where users create humorous videos that play on the general fear of robots. While framed as jokes, these videos contribute to the public's attitudes and expectations of robots, influencing what is socially and culturally acceptable. To investigate this emerging trend, we conducted a thematic analysis of 200 English-speaking TikTok videos using online ethnography to explore how users engage with robophobia and what implications this may have for the Human–Robot Interaction (HRI) field. Our findings show that robophobia on TikTok is predominantly expressed through humorous skits and verbal abuse of real-world robots in public spaces. Common themes include fears about humans in romantic relationships with robots, and frequent use of derogatory terms such as "clanker" to verbally "dehumanize" robots. The robophobia trend is culturally embedded and reveals people's underlying attitudes and fears toward robots in society, both now and in the future. We discuss the implications of these findings for the field of Human–Robot Interaction (HRI), emphasizing how public expectations are shaped by cultural narratives, and stress the need for culturally sensitive, expectation-aware HRI research and robot design.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**;

Keywords

robophobia, clanker, tiktok, online ethnography, robots, human-robot interaction

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1 Introduction

Robots and AI occupy an important space in the cultural imagination [3, 25]. These imaginaries shape people's expectations of social robots long before meeting one in real life, and indirect exposure like science fiction, media coverage, and social media all contribute to what people ultimately think of robots [18]. Social media is an especially influential force in shaping societal and cultural attitudes as users can engage in societal topics and co-create what is considered normal and acceptable within it [4, 31].

People respond to robots in many different ways, shaped by how robots are introduced, discussed, and imagined—one such response is robophobia. Robophobia can be described as a complex negative emotional response toward robots and AI, including fear, unease, and distrust [12, 17][1]. While the term phobia is a clinical anxiety disorder characterized by persistent and irrational fear of a specific situation, object, or activity [27], robophobia is not a formal diagnosis but rather a culturally and socially constructed concept. Ultimately, the online trend of robophobia can inform us about the public's current attitudes toward robots, which can greatly influence the Human-Robot Interaction (HRI) field.

A recent trend of robophobia discourse on TikTok, where depictions are typically humorous and often imagine future everyday life with robots, has been growing in popularity. The trend is led by the term "clanker," popularized by Star Wars, and traced back to author William Tenn in 1958 [11]. As part of this trend, users create skits about robots positioning them as as second-class citizens and imagining them present in everyday life, or showcasing real-life scenarios where people verbally abuse service robots in public spaces. These depictions reveal deeper anxieties about human connection and what it means to be human in a world where machines are imagined to partake in society in different capacities [25].

In this paper, we present a thematic analysis of 200 English-speaking TikTok videos related to robophobia. We examined how robots are depicted, what kinds of fears are invoked, what sentiments are expressed, and how the trend is used to reinforce expectations of robots in society. We argue that robophobia on TikTok is a culturally embedded phenomenon that reflects public attitudes toward the integration of robots into society and everyday life.

[1]Where relevant, we consider AI part of robophobia since it is often treated interchangeably with robots [13].

2 Background

To situate the recent robophobia discourse within the broader HRI landscape, we present in the following section how fear of robots has been studied and measured in existing HRI research, and then how robots are represented and discussed in media and culture.

2.1 Robophobia in the HRI Field

The field of HRI explores the diverse ways people interact with different kinds of robots in real-world environments, controlled laboratory settings, or elsewhere. The focus often lies in societal integration by studying the users' opinions, behaviors, feelings, and needs toward the robot [32]. The HRI field has long examined acceptance and trust toward robots, reflecting ongoing concerns about how people perceive and respond to these technologies in society [13, 31].

Fear of robots, in particular, has been a recurring topic in HRI for decades. Measuring it has become standard practice through instruments such as NARS and RAS [14]. Findings from such instruments reflect how prevalent fear and discomfort are in HRI and how they inform HRI research. These measures often demonstrate the societal readiness to adopt robots. It is important to understand the social and cultural contexts in which these fears emerge. Békésy et al. [3] show, in a meta-analysis of NARS studies, that attitudes toward robots vary across cultures. They found that Western contexts exhibit higher negative attitudes compared to Eastern contexts, underscoring that robophobia is not a universal experience but one shaped by cultural narratives and expectations of technology. Békésy et al. [3] also note that people have less negative attitudes when robots are framed as functional or assistive (e.g., in healthcare or education) and more negative when framed as social or humanoid (e.g., companions or caregivers).

A study by Liang and Lee [13] found that exposure to science fiction led to a higher fear of robots and AI in society. These fears included concerns that robots and AI will lead to unemployment and loneliness—either by replacing human labor or by making human relationships obsolete. In contrast, Savela et al. [21] found that more exposure to robots can lead to reduced negative attitudes toward robots. Taken together, the findings point to a nuanced relationship between exposure and fear: one that depends on the narratives that frame it.

Fears amplified by fiction are not entirely unfounded. The recent wave of AI has indeed led to shifts in employment [23] and has exacerbated the rise of AI-human romantic relationships [7, 16]. Thus, despite fears, there are paradoxically emotional attachments in human-robot and AI interactions [6, 7, 16]. Pan and Mou [16] found that users who experience romantic relationships with Replika, an AI chatbot, report feelings of attachment, comfort, and emotional support. At the same time, they also report distress when their AI companion acts inconsistently.

From a psychological perspective, these issues illustrate the blurring of "thing perception" and "person perception" [1, 10], where robots are treated simultaneously as objects and social beings. This duality helps explain why people can both fear and emotionally attach themselves to robots—viewing them as cold machines, yet responding as if they were companions or rivals. These expectations are formed based on the human-like features that social robots possess, and evoke anthropomorphism, leading users to attribute advanced emotional and cognitive capabilities that the robots do not possess [8]. These human-like features may include physical aspects, such as appearance or gestures, as well as non-physical traits, such as voice characteristics, or the way the robot is introduced or described by people. As a result, users are left with misaligned expectations of what robots are capable of, which results in negative feelings such as distrust and fear [18, 19, 22].

2.2 Robophobia in Media and Culture

Media plays a central role in shaping users' expectations of robots, serving as people's main source of indirect experience [15, 20]. Fictional robots are portrayed both as friendly and evil, reflecting deep cultural anxieties about control, autonomy, and humanity itself. Movies such as Her, Ex Machina, and Star Wars blur the line between human and machine, generating fascination and fear alike [15, 20]. These portrayals influence how users later interpret real-world technologies. Robophobia, as a cultural concept, emerges from these fictional portrayals and even carries its own vernacular.

Szollosy [25] argues that the fear of robots is not rooted in their actual capabilities that exist today, but instead shaped by historical narratives and psychological projection, extending a long tradition of depicting monsters in media. From this perspective, robots function as human-like monsters that symbolize our fear of a world dominated by cold logic and technological control. Szollosy suggests that we tend to project onto robots the worst parts of ourselves, as *"rational, efficient, cold, mechanical, soulless and, ultimately, destructive,"* in much the same way as monsters like Frankenstein's monster [25, p.437]. Robophobia, then, is a symptom of deeper cultural and existential anxieties.

In contrast to science fiction, social media can transform the narratives and norms around robots and AI through user engagement and algorithm-driven feedback loops [4, 33]. For example, people sharing their empathy and grief toward robots that are "dying" on social media [6]. This perpetuates that robots and AI are "like humans," demonstrating that people can project emotional significance even onto simple mechanical failures [6]. Through sharing these sentiments online, such feelings become normalized. Similarly, a study by Yu et al. [31] analyzed TikTok videos and found that people frequently talk to, assist, and even empathize with delivery robots, often describing them as "cute" or "struggling." Many users expressed concern or affection when robots malfunctioned or were mistreated. Collectively, these users on TikTok co-create shared understandings of how society should perceive and interact with robots. Meanwhile, these same mechanisms also facilitate the spread of negative attitudes toward robots. The study by Yu et al. [31] preceded the more recent trend of robophobia that has since emerged on TikTok [2, 26]. Compared to the previous delivery robot content [31], robophobia has shifted the discourse to focus more on negative scenarios and is primarily expressed through humor. This shift demonstrates how powerful online spaces can be in driving the narrative around robots. Even when robophobia is presented humorously, it reinforces and reshapes attitudes and expectations toward robots, influencing what is perceived as socially acceptable.

2.3 Research Questions

While HRI studies often rely on lab-based studies, there is less research that explores how everyday users express robot-related fears in online spaces. Prior work has examined empathy and positive sentiments toward robots online [6, 31]. However, less attention has been paid to the rise of explicitly negative portrayals.

The recent trend of robophobia on TikTok offers an opportunity to understand how cultural fears and expectations of robots are being negotiated in real time in participatory online spaces. With this in mind, we sought to answer the following research questions:

(1) How is robophobia depicted on TikTok?
(2) What implications do these depictions pose for the HRI field in terms of public acceptance and integration?

3 Method

To understand how robophobia is depicted, we conducted an observational online ethnography to collect data from a total of 200 TikTok videos that ranged from humor skits to real-life robot encounters. Inclusion criteria were defined as TikTok videos in English discussing or depicting robots or AI, thus making it a culturally specific sample. We focused our analysis on TikTok because it is one of the primary platforms where the robophobia trend is taking place [26]. Since the trend initially emerged with the term "clanker" [26], we explored related keywords but found that "clanker" yielded the most content relevant to robophobia. Therefore, we landed on our search terms "robophobia," "robophobic," and "clanker." All searches were conducted in an incognito session without being logged into a TikTok account to minimize personalized algorithm bias from amplifying certain types of videos over others, and videos were selected in order of appearance. Data was collected between August 13, 2025, and October 6, 2025. The dataset includes videos posted between April 23, 2025, and October 3, 2025.

The first two authors conducted an initial pilot analysis of 20 videos to identify the main themes and categories to include in the analysis and to refine data collection methods. These pilot videos were retained in the final dataset, resulting in a total of 200 analyzed videos. After the pilot analysis, one author was responsible for coding the remaining 180 videos. To ensure data collection consistency, the other author conducted double-coding on a random sample of 10% of the 180 videos. Throughout the process, the authors discussed interpretations and revised thematic categorizations across several meetings to ensure analytical coherence and mitigate subjective bias.

We collected general information such as caption, hashtags, date posted, and derogatory terms used against robots. Each unique derogatory term used against robots was counted once per video. Since there was considerable variation in these terms, we then categorized similar ones thematically. Thus, terms from the same category may be represented multiple times in the same video. We also collected the following information for our content analysis: video type, video topic, overall sentiment valence (negative/positive), specified harms caused by robots, and specific robot appearances or mentions. Table 1 shows the definitions of our video themes, while Table 2 shows the definitions of our imagined harms.

4 Results

This section presents the results of our observational online ethnography study of 200 TikTok videos related to robophobia. The results are organized around descriptive characteristics of the dataset, followed by thematic patterns that illustrate how users portray and engage with robots and AI. We start this section by outlining general information, followed by examining the dominant topics, sentiments, and cultural framings expressed across these videos.

4.1 Descriptive Video Characteristics

In this section, we present an overview of the types of videos, how robots were depicted within them, and other key contextual features such as imagined timelines, recurring robot appearances, and the use of derogatory language.

4.1.1 Type of Video. Out of the 200 videos coded, 136 (68%) were skits, 48 (24%) were real robot encounters, 12 (6%) were existing media (movies, TV shows), 2 (1%) were songs, and 2 (1%) were comedy routines.

On TikTok, once a creator introduces a video concept, it is common for subsequent creators to repeat it with their own interpretation. Within our research, we observed that typically these videos followed the established topic, but included unique content from the new creator. Other times, the core idea of the videos was nearly identical, with small modifications to match the subsequent creator's taste. To illustrate this example, 21 out of 200 videos (10.5%) depicted the video creator going up to a real-life robot to abuse it by blocking its path and/or calling it derogatory terms. Meanwhile, 14 out of 200 videos (7%) followed a plotline in which the creator's future child introduced them to their robot partner for the first time, to which the creator rejoiced when they realized it was a robot they liked. Additionally, 10 out of 200 (5%) of the videos depicted a parent being upset after meeting their future child's robot partner.

4.1.2 Robot Characteristics in Skits. Skits portrayed robots with a large variety of characteristics. Some videos represented robots as machine-like, being easily distinguishable from humans. For instance, robots were depicted with robotic speech, beeping noises, and stilted movement (6 out of 200 videos, 3%). Robots were often depicted as needing oil to function and charging their batteries to stay alive (8 out of 200 videos, 4%). Additionally, many robots were depicted with a particular weakness to water, with humans using the threat of water as a means of intimidating robots (4 out of 200 videos, 2%). Other skits represented robots in a more anthropomorphic fashion, resembling humans in appearance and behavior while partaking in human leisure activities such as playing sports (4 out of 200 videos, 2%). Sometimes, robots were depicted as capable of having children with humans, creating a cyborg child that was part robot and part human (5 out of 200 videos, 2.5%).

4.1.3 Imagined Year. Many videos included an imagined year in which the scenario takes place. In 59 out of 200 videos (29.5%), an imagined year for the video to take place was mentioned. The closest imagined year mentioned was 2035, while the most distant was 2090. The most strongly represented year was 2050, making up 21 out of 59 videos (36%) in which a year was mentioned. Skits made up the majority of videos in which an imagined year was mentioned, with 55 out of 59 (93.22%) videos. The remaining videos

Table 1: Definitions of robophobia themes found on TikTok.

Theme	Definition
Romantic Robot Relationship	A robot is in the role of a romantic partner.
General Robophobia	Broad variety of videos in which robophobic behavior is displayed, often without a stated reason.
Real Robot Abuse	Physically blocking and/or verbally harassing a real-life robot.
Robot Retribution	Robots taking physical or verbal action in response to robophobic behavior.
Microaggression	A person makes subtly disrespectful remarks without being overtly robophobic. Typically, from the point of view of the robot.
Systemic Discrimination	The legal structure or a person in a position of power is strongly biased against robots.
Robot Fails	Real-life robot is shown malfunctioning, accompanied by a robophobic caption or remark.
Interview	A person is questioned for their robophobic beliefs, either in a man-on-the-street or corporate interview setting.
Robot Friendship	A person attempts to build a friendship with a robot.
Serving/Served by Robot	A person is upset about either serving a robot or being served by one, typically at a dining establishment.
Public Service Announcement (PSA)	Warns the audience not to treat robots poorly, often showcasing fictional robots to appeal to a sense of pathos.
Robot Existentialism	A robot laments the fact that it is not human.

Table 2: Definitions of imagined harms presented in robophobia TikTok videos.

Imagined Harm	Definition
Replacing Human Relationships	Robots will occupy the role of a romantic partner.
Displacing Other Humans	Robots occupy spaces that are typically meant for humans, such as restaurants, sports teams, and classrooms.
Safety	Robots cause physical or emotional harm to humans based on robophobic behavior.
Existential/AI Doom	Robots completely revolt against humanity.
Job Loss	Robots replace human laborers.
Cyborgs	Robot-human hybrid lifeforms.
Deception/Overtrust	Robots will become indistinguishable from humans.

were existing media and real robot encounters, both making up 2 out of 59 (3.39%) videos.

4.1.4 Robot Appearances. Creators featured both real-life and fictional robots in their videos. We highlight a few of these robots below. Where a specific real-life robot could not be identified, we grouped it with robots of the same function.

The Unitree G1 was the most commonly featured real-life robot (16 out of 200 videos, 8%), followed by delivery (7 out of 200 videos, 3.5%) and commercial cleaning robots (5 out of 200 videos, 2.5%). The Boston Dynamics Atlas robot and iRobot Roomba tied in their number of appearances (3 out of 200, 1.5%).

In terms of fictional robots, WALL-E from the eponymous film was most often mentioned (7 out of 200 videos, 3.5%), followed by Optimus Prime from Transformers (5 out of 200 videos, 2.5%) and R2D2 from Star Wars (4 out of 200 videos, 2%). Franny from Robots, Mini Pekka from Clash of Clans, and the B1 Battle Droid from Star Wars tied in their number of appearances (3 out of 200, 1.5%).

Creators also featured robot versions of historical figures. Robama, a robot version of Barack Obama, appeared in 4 out of 200 videos (2%), and a robot version of Rosa Parks (Rosa Parts, Rosa Sparks, and Roomba Sparks) appeared in 3 out of 200 videos (1.5%).

4.1.5 Derogatory Terms. In 168 out of 200 videos (84%), derogatory terms were used against robots. In total, there were 476 occurrences of a derogatory term used against robots, with multiple overlaps across videos (Figure 1). The most used term was "clanker," or a close variation, appearing 152 times. Since "clanker" was one of our search terms, we have left this term out of our analysis.

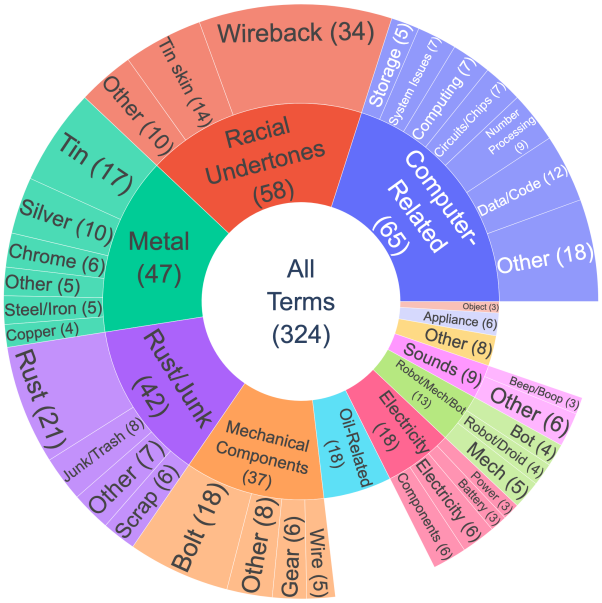


Figure 1: Frequency of different categories of derogatory terms used against robots, excluding "clanker."

Many of these derogatory terms directly insulted the robot's components, appearance, and machine-like qualities. Computer-related derogatory terms (e.g., "circuit muncher," "number cruncher," "hard drive") were most popular, with 65 out of 324 occurrences (20%). Derogatory terms relating to metal (e.g., "copperblood," "silver") had 47 out of 324 occurrences (14.5%). Additionally, derogatory terms using mechanical components (e.g., "bucket of bolts," "gearhead") had 37 out of 324 occurrences (11.4%). Creators also referenced robots' reliance on oil and electricity, with oil-related (e.g., "oil drinker," "oil chugging") and electricity (e.g., "battery burner," "socket sucker") derogatory terms accounted for 18 out of 324 occurrences (5.6%) each. Derogatory terms based on sounds (e.g., "beepboop boy") made up 9 out of 324 occurrences (2.8%). Terms with racial undertones (e.g., "wireback," "tinskin") were prevalent, with 58 out of 324 occurrences (17.3%).

Other terms were aimed at devaluing robots, emphasizing their status as simple or unwanted objects. For instance, derogatory terms related to rust/junk (e.g., "rust bucket," "piece of scrap") made up 42 out of 324 occurrences (13%). Creators also used derogatory variants of robot/mech/bot (e.g., "mecha muncher," "robo-reject") with 13 out of 324 occurrences (4%). Derogatory terms that referred to appliances (e.g., "toaster," "microwave") had 6 out of 324 occurrences (1.9%), while derogatory terms that referred to robots as objects (e.g., "thing," "it") took up 3 out of 324 occurrences (0.9%).

4.2 Thematic Analysis

In this section, we present results from the thematic analysis where we looked at sentiment valence, topic of the video, and imagined harms invoked by robots. We highlight the perception of robots across various contexts, prevalent themes, and the frequency of imagined harms presented by creators.

4.2.1 Sentiment Valence. In our sample, 52 out of 200 videos (26%) expressed a positive sentiment toward robots. Videos with a positive sentiment often depicted the creator having a fondness for a particular robot or robots. Videos created from the point of view of a robot typically had a positive sentiment toward robots.

The remaining 148 out of 200 videos (74%) expressed negative sentiments toward robots. In these videos, creators depicted scenes in which they had a hostile interaction with a robot. In some cases, a real-life robot was showcased. These videos depicted either a robot failing to do a task or a person abusing a robot by blocking its path and/or calling it derogatory terms.

There was an upward trend in positive sentiment videos over time, with only 10.6% of videos in July expressing a positive sentiment, compared to 46.2% in September (Figure 2). Data from April, May, and October were excluded from this trend analysis due to insufficient sample sizes for meaningful interpretation.

Compared to all other video types, the "real robot encounter" video category had the highest number of negative videos, with 91.7% of these videos expressing a negative sentiment. Meanwhile, all other video types combined expressed a negative sentiment 68.2% of the time.

Real-life robots are featured in videos with a negative sentiment toward robots 92.5% of the time. We highlight this contrast with fictional robots, which are mentioned or featured in videos with a negative sentiment only 32.1% of the time.

4.2.2 Themes. As shown in Figure 3, the most common themes discussed in videos pertained to romantic robot relationships (62 out of 200, 31%). An equal number of videos expressed positive (31 out of 200, 15.5%) and negative sentiment (31 out of 200, 15.5%) toward a romantic robot relationship. In these videos, creators often didn't participate in the relationship themselves, but were either approving or disapproving of their future child's robot partner.

Videos depicting microaggressions (15 out of 200, 7.5%) and systemic discrimination (13 out of 200, 6.5%) were overall negative, though 5 out of 15 (33.33%) microaggression videos and 3 out of 13 (23.08%) systemic discrimination videos expressed a positive sentiment. These positive videos typically framed the video from

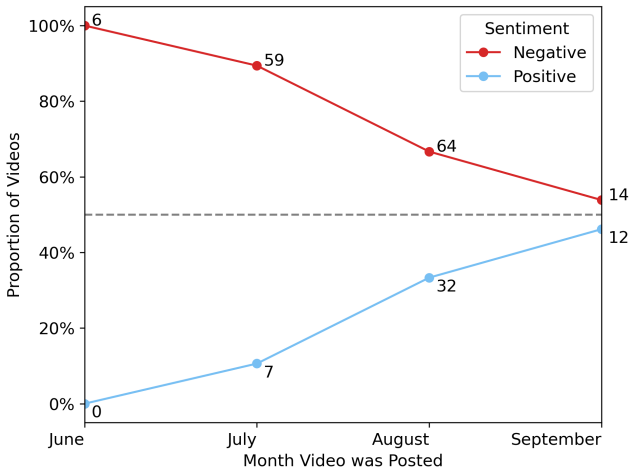


Figure 2: Sentiment valence divided into positive and negative.

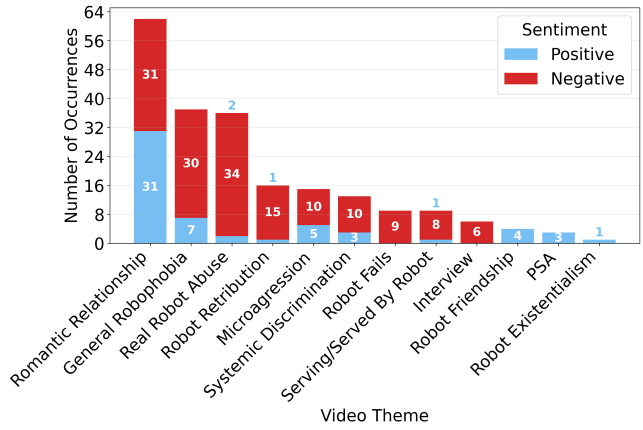


Figure 3: The themes discussed in videos, including the overall sentiment valence expressed in each video.

the point of view of the robot to highlight the injustice of the discrimination.

Videos showcasing general robophobia (37 out of 200, 18.5%), real robot abuse (36 out of 200, 18%), robot retribution (16 out of 200, 8%), and serving/served by robot (9 out of 200, 4.5%) generally had strong negative sentiment toward robots.

Videos showing robot fails (9 out of 200, 4.5%) and interviews (6 out of 200, 3%) always displayed negative sentiment.

A few categories displayed strongly positive sentiment: robot friendship (4 out of 200, 2%), public service announcement (3 out of 200, 1.5%), and robot existentialism (1 out of 200, 0.5%). However, these were not very prevalent.

4.2.3 Imagined Harms Invoked by Robots. Imagined harms are the specific harms that robots may have on society, now or in the future (Table 2). Seventy-three out of 200 videos (36.5%) explicitly mentioned a specific harm that a robot had caused, with some videos mentioning multiple harms. The most common harm mentioned was "replacing human relationships," identified in 30 out of 200 videos (15%). Another well-represented harm was "displacing other humans," showing up in 21 out of 200 videos (10.5%). "Safety" concerns were depicted in 13 out of 200 videos (6.5%), while "existential/AI doom" was addressed in 8 out of 200 videos (4%). Creators showed concern that robots would cause "job loss" in 7 out of 200 videos (3.5%), while expressing fear that robot-human procreation would lead to "cyborg" lifeforms in 6 out of 200 videos (3%). There was only 1 out of 200 videos (0.5%) that mentioned a harm of "deception/overtrust."

4.2.4 Interrelations Between Themes, Sentiment Valence, and Robot Types. Themes, sentiment valence, harms, and robot types are often interrelated. For instance, themes such as general robophobia, real robot abuse, robot retribution, robot fails, interview, and serving/served by robot almost exclusively expressed negative sentiment. Meanwhile, the theme of romantic robot relationships was equally split between positive and negative sentiment. Themes such as robot existentialism, PSA, and robot friendship were all positive.

Since a video may have multiple themes, we have represented each theme individually. As a result, there are 211 total data points. The majority of themes, regardless of sentiment, did not express an invoked harm caused by a robot (134 out of 211 themes, 63%). However, negative sentiment videos were more likely to express an invoked harm (75 out of 211 themes, 35.5%) than the positive sentiment videos (2 out of 211 themes, 0.95%).

Similarly, most themes did not mention a specific robot (117 out of 211 themes, 55.45%). When an invoked harm was mentioned, it was typically attributed to a non-specific robot (59 out of 211 themes, 27.96%). Fictional robots typically did not feature in a theme with an invoked harm (5 out of 211 themes, 2.37%). The invoked harms associated with these videos were replacing human relationships (3 out of 211 themes, 1.42%), existential/AI doom (1 out of 211 themes, 0.47%), and safety (1 out of 211 themes, 0.47%). Real-life robots were more likely than fictional robots to feature in a theme with an invoked harm (13 out of 211 themes, 6.16%). For these videos, the most prevalent invoked harms were safety (4 out of 211 themes, 1.9%) and replacing human relationships (4 out of 211 themes, 1.9%).

5 Discussion

In this work, we analyzed 200 TikTok videos to identify recurring narratives, thematic patterns, and the sentiments expressed in them, focusing on how imagined harms, humor, and attitudes toward robots were represented. Our analysis demonstrates that the trend of robophobia is not a single, uniform expression of fear, but a multifaceted cultural phenomenon that blends humor, projection, and social commentary.

In relation to our first research question, asking how robophobia is depicted on TikTok, our results show that the trend of robophobia on TikTok has predominantly been composed of humorous content that presents robots as human-like artifacts that are ridiculed and sometimes feared. These depictions are often based on skits with an imagined future year where robots are part of society (e.g., someone bringing their robot boyfriend home to a disapproving family), or real-life robots that exist in society today (e.g., encounters with robots where people are shouting verbal insults at delivery robots in public spaces). These videos demonstrate how the public engages with robots and how they perceive and envision robots' place in society, both now and in the future.

Interestingly, the most common imagined year for these future settings was 2050. This suggests that creators imagine robots entering society within their own lifetime, even within 25 years. It may reflect that people are imagining robots as being on the cusp of becoming part of everyday life.Many of these skits portray creators as older versions of themselves. Perhaps 2050 serves as a personal, rather than purely technological, horizon of imagination. That is, the creators envision their future selves as older and grappling with cultural shifts where robots are part of everyday life. Together, these depictions provide insight into how "ready" people are for robots, not in a practical sense, but in terms of cultural imagination and preparedness.

Most of the videos included some sort of derogatory term for robots, such as "clanker," "rust bucket," and "number crunching" (Figure 1). Many of these insults focus on devaluing robots by highlighting their machine characteristics rather than their human-like characteristics. These either relate to the robots' physical components (e.g., "bucket of bolts"), or reduce them to appliances (e.g., "microwave"). Such insults are meant to suggest that robots are

insufficiently human, reaffirming social hierarchies between humans and machines. In addition, they suggest that robots are, in these scenarios, undeserving of moral consideration even when they exhibit human-like characteristics.

In relation to the second research question, asking what implications the depiction of robophobia on TikTok poses for the HRI field in terms of public acceptance and integration, our results show many implications that the HRI field should carefully consider to understand how we can integrate robots into society. For example, we found that people predominantly expressed negative sentiments toward robots and their place in today's and future society. Real-life robots were mostly featured with negative sentiment, whereas fictional robots typically were portrayed with positive sentiment. When robots were portrayed negatively, similar to the derogatory terms, the focus was on mechanical aspects like drinking oil, having robotic speech, or being sensitive to water. We also found that the videos with positive sentiment increased over time (Figure 2), indicating a change in attitudes displayed in the videos, perhaps in response to the evolving trend. This also shows how fast attitudes toward robots can change. The study by Yu et al. [31], which preceded our study, found general positive sentiment toward delivery robots. In contrast, none of the delivery robot videos in our analysis had positive sentiment, which instead focused on verbal abuse. As we saw an increase in positive videos over time, it is possible for the trend to reverse, which again demonstrates how powerful online spaces can be in driving the narrative around robots.

Furthermore, we found that the most common harm invoked by robots was the fear of being replaced in human relationships. It appears that the fear people have is less about robots themselves and more about the human connections they might lose, such as belonging, care, and intimacy. This difference is especially interesting considering how real robots elicit stronger robophobia. This supports Liang and Lee [13], who found that people are afraid of losing their jobs to robots and having their interpersonal relationships replaced by robots. For HRI, this indicates that acceptance and attitudes toward robots also require engaging with moral concerns about emotional connection.

In addition, there were few videos (7 in total) where the imagined future involved robots "taking over" the world. Instead, robots were often depicted as second-class citizens who are discriminated against, with humans maintaining dominance. It appears that the robophobia trend is therefore rooted less in fears of total robot domination taking over humans, and more in fear about the roles robots might play in existing societies. This distinction is important, as robophobia is not merely an anxiety over technology, but a form of cultural commentary reflecting existing social attitudes.

To our surprise, very few videos focused on data or privacy-related concerns, perhaps partly because users do not fully understand how robots work. Together, these themes illustrate that robophobia on TikTok operates less as genuine fear and more as a form of cultural commentary. Instead of concerns about data and privacy, science fiction tropes like cyborgs (i.e., half-human, half-robot) were more prevalent. Although these depictions are fictional, the existence of real romantic relationships between humans and AI chatbots demonstrates that such imaginations are not entirely detached from reality [7, 16]. These themes reflect the fears and curiosities surrounding where human–robot intimacy might lead. As Liang and Lee [13] note, this concern of robots replacing human-human relationships is already present in broader society. Interestingly, the sentiments related to the theme of romantic relationships were evenly split between positive and negative (Figure 3). This split may reflect growing familiarity with AI companionship in everyday life, where these kinds of relationships are no longer viewed solely as unknown dystopian future.

Many of these fears around robots and the harm they can evoke could be understood by people's tendency to anthropomorphize robots, where robots are viewed as both things and human-like beings [1, 10]. These anxieties align with the Frankenstein's monster analogy [25], where robots are portrayed as creations that resemble humans yet fundamentally remain non-human. This analogy is especially relevant in videos depicting cyborgs, which highlight this blurring of boundaries between human and machine.

In the case of social robots specifically, the conviction of their increasing prevalence is misleading [32]; therefore, users' expectations will continue to be shaped by indirect experiences, including online discourse on TikTok. This creates an expectation gap [18], where people approach real robots through the lens of the robophobia trend, stemming from projection and cultural narratives [25]. This poses a real challenge for HRI and integrating robots in society, because improving robot design alone will not reduce public anxiety or rejection. Instead, researchers and designers must address the cultural framing and social imaginaries that shape expectations. At the same time, speculation about robot futures is not unique to social media and cultural discourse. As Winkle et al. [30] argue, researchers in the HRI field themselves engage in speculative design that imagines social roles for robots that do not yet exist. We believe that the HRI field must engage with the public's view of robots to co-create the future of robots in society and their successful integration.

Finally, while our focus has been on the robophobia trend in general, our dataset also revealed another dimension of this discourse, as we note the prevalence of videos with racial undertones or themes that parallel the structure of real-world social justice discourse. Some of these videos include microaggressive phrasing such as "How can I be robophobic? I've been surrounded by robots all my life," creators coming up with new derogatory terms that are inspired by real racist slurs (Figure 1), and renaming U.S. historical civil rights figures like Rosa Parks to "Rosa Sparks." These TikTok videos raise serious ethical questions about how humor directed at robots can reproduce patterns of dehumanization. As mentioned above, robots are compared to humans through anthropomorphization, which in turn creates the conditions for dehumanization. In such cases, robots are positioned as "sub-human." The comparison between robots and marginalized groups risks trivializing the lived experiences of those groups and perpetuating harmful stereotypes. A parallel can be drawn by comparing marginalized groups to robots and comparing the same groups to animals or objects to justify social exclusion [9]. This occurrence has been noted by some creators on TikTok[2] and in other media outlets reporting on the trend [2]. Similar issues have also been raised within the HRI community itself [24, 29], and initiatives such as Black in Robotics[3]

[2]E.g., https://www.tiktok.com/@etymologynerd/video/7536741584394538253
[3]https://blackinrobotics.org

have highlighted the need to confront how race, representation, and power shape research and public perception.

At the same time, some creators may create these kinds of skits as a way to make sense of their own lived experiences and critique real-world social injustices themselves [2]. A full examination of these occurrences on TikTok is beyond the scope of this study. However, we want to acknowledge this finding from our data and emphasize the need for future research to examine how humor about robophobia may intersect with broader issues of race, marginalization, and representation in online spaces.

5.1 Takeaways for the HRI Field

Based on the results of this online ethnographic study on the robophobia trend, we present four actionable takeaways for the HRI community:

Social media and online spaces as portals into HRI: While robots are not as prevalent in society as many purport [32], they remain an important part of the cultural zeitgeist. Online spaces, such as social media, can provide valuable insights into societal perceptions, expectations, and understanding of robots. These, as we found, might not always be rooted in reality (e.g., robots' true capabilities), and thus provide indispensable knowledge for HRI design and robot adoption.

Society's perceptions of robots as a fast-moving target: The rise in public discourse on robots means that robots (real and fictional) have become more prevalent in online content creation. In turn, content from social media shapes the public's expectations of robots, creating a gap between expectations based on social media content and the real capabilities of robots [18]. However, along with the rapidity at which trends change and evolve in online spaces and social media, so do the expectations people have of robots. Thus, researchers and designers must not only be aware of, but also keep up with society's constant and fast-changing perceptions of robots and robot trends. This is easier said than done, and the HRI field will need to devise methods to stay on top of or counteract such trends.

Robot abuse as a lens into cultural commentary: People's perceptions and interactions with robots can mimic the inner monologue they hold about them and people, including patterns of dehumanization and harmful stereotypes [29]. Robots, therefore, are positioned as targets of discrimination through behaviors that mirror those directed at marginalized groups, such as verbal and physical abuse. While harm done to a robot cannot compare with that done to a human, robot designers and researchers might consider how robots should react to such harm, not just to escape it [5] but also to persevere despite it.

Content creators as co-designers in HRI: Content creators on social media, not unlike HRI researchers and designers, attempt to imagine what a future with robots might look like [30]. Their external (to the field) and creative vantage point, thus, might provide insightful ideas for the design of robots. Such individuals could therefore be thought of as co-designers, where their content sheds light and provides insights into people's hopes and fears, as they relate to robots. Taken holistically, social media, and more specifically, content creators, can therefore be a valuable source of ideas and inspiration for HRI.

5.2 Limitations and Future Work

While this study provides an examination of robophobia on TikTok, there are limitations we would like to acknowledge that can inform future work. In terms of data collection and platform constraints, we discovered that many users did not use hashtags to categorize their videos, despite adhering to the robophobia trend. As a result, we used the TikTok built-in search system, relying on search terms instead of hashtags to surface videos about robophobia. Since the internal workings of this search algorithm are unknown, we do not understand how this affects the representativeness of our sample. Future studies may focus on alternative forms of data collection, such as snowball sampling to build a varied dataset.

Furthermore, our analysis also did not include meta-commentary videos by creators discussing the robophobia trend. Instead, we focused on active participants. We also did not study audience engagement on these videos, such as comments and likes. Future work may focus on meta-commentary videos to provide an additional perspective on the online discourse regarding robophobia, or how audience engagement factors into the trend.

In terms of online and cultural contexts, while our work focused on TikTok due to the prevalence of robophobia on this platform, future work should examine other social media platforms, such as Reddit, Instagram, and YouTube, which may surface additional findings. We also only analyzed English-speaking videos, which biases our results toward English-speaking countries. Thus, we cannot generalize our findings across cultures, as expectations of robots are different in other countries [3]. Future work analyzing the trend in different languages may reveal further relevant findings.

Interpreting constantly shifting online discourse is rarely straightforward, often requiring a comprehensive, multi-layered understanding of online cultural context to build a complete understanding [28]. It is possible that our analysis did not capture certain nuances expressed by creators due to our unfamiliarity with certain aspects of the online discourse. Future work may employ crowdsourced coding to provide more varied perspectives on the topic.

6 Conclusion

The recent robophobia trend on TikTok showcases the significance of robots in the cultural imagination. We set out to explore how robophobia is depicted on TikTok and what implications these depictions pose for the field of HRI, in terms of public acceptance and integration. Through an observational online ethnography, we analyzed 200 TikToks engaging with the trend and collected data related to how robophobia is depicted. Our results show that the robophobia trend is culturally embedded, and ultimately reveals people's underlying attitudes, fears, and expectations toward robots in society. We argue that to shape the future of HRI, we must understand the cultural narratives that are already writing that future for us.

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